Roll No. | | | | | | | | | | |

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.-7)

INSTRUMENTAL METHODS OF ANALYSIS-THEORY

Subject Code : BP-701T

M.Code : 78387

Date of Examination :21-11-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Write briefly :**

   a) How does solvent polarity affect absorption maxima?

   b) What is fluorescence quenching?

   c) What are the fundamental modes of vibration in polyatomic molecules?

   d) Give one application of flame photometry in clinical or pharmaceutical analysis.

   e) Distinguish briefly between nephelometry and turbidimetry.

   f) Differentiate between line source and continuum source in atomic absorption spectroscopy.

   g) Write any two differences between Paper Chromatography and Thin Layer Chromatography (TLC).

   h) Differentiate between adsorption and partition chromatography in terms of mechanism.

   i) Define retention time and explain its significance.

   j) Define cation-exchange resin and give an example.

(S15)- 301

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SECTION-B

2. Explain the factors affecting vibrational frequencies in IR spectroscopy. Additionally explain the working principle of Golay cell.

3. Describe the instrumentation of HPLC, including pump, injector, column, and detector.

4. Explain the principle, construction, and working of a photomultiplier tube.

SECTION-C

5. Derive the Beer-Lambert's law and explain possible deviations.

6. Explain the theory of fluorimetry and the mechanism of fluorescence.

7. Discuss the applications of flame photometry in quantitative analysis of metals and clinical samples.

8. Discuss factors affecting nepheloturbidometric measurements.

9. Discuss Rf values in TLC, their calculation, significance and factors affecting them.

10. Discuss the factors affecting electrophoretic mobility, including charge, size, shape and buffer composition, with examples.

11. Compare GC and HPLC in terms of separation mechanism and sensitivity.

12. Describe the properties of ion exchange resins, such as capacity, selectivity and physical characteristics.

13. Describe the types of ligands used in affinity chromatography and their applications.

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.

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